Software Requirement Specification Hostel Management

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Project ID	16
Problem Statement	Hostel Management

1. Introduction

1.1. Purpose:

The purpose of this document is to outline the specifications for the development of a comprehensive hostel management portal. This portal will facilitate efficient management of room capacities, occupancy statuses, Outsiders room status communication between administrators and students. for both girls and boys. It will provide administrators with tools to effectively allocate resources, track room vacancies, and oversee hostel operations and informations.

1.2. Scope of Project:

- The software system aims to streamline hostel management processes by providing administrators with a centralized platform for monitoring and managing room

capacities, occupancy statuses, related data and enable students to report hostel-related issues..

- Users will include hostel administrators responsible for overseeing room allocations, as well as staff members involved in the management of hostel facilities.
- The system will offer features such as real-time updates on room vacancies, detailed room allocation records, and insights into hostel infrastructure metrics.

2. System Overview:

2.1. Users:

1. Administrators: Responsible for managing room capacities, overseeing room allocations, and accessing analytical insights related to hostel operations.
2. Staff Members: Assist administrators in day-to-day hostel management tasks and contribute to the upkeep of hostel facilities.
3. Students: Utilize the system for room booking, reporting issues, and accessing relevant information.

2.2. Features:

1. Room Capacity Management: Administrators can oversee and adjust room capacities based on organizational requirements.
2. Room Vacancy Status Tracking: Provides detailed insights into the occupancy status of every room within the hostel premises which allows us to track vacant rooms.
3. Room Allocation Details: Comprehensive records of room allocations, including

occupant details and allocation history.

4. Type of Room Information: Catalog detailing different types of rooms available in the hostel, along with specifications and amenities.
5. Hostel Block Details: Overview of hostel infrastructure, including block layouts, facilities, and floor plans.
6. Outsider Room Booking: Portal for outsiders to request room bookings, with features for approval and payment processing.
7. Total Admitted Strength: Analytics dashboard displaying historical trends and projections regarding hostel admissions.
8. User Complaint System: Platform for students to report hostel-related issues, with administrators having the ability to respond and address concerns.
9. Biometric Register Details: Implementation of a system to record biometric data of students regularly (every day from 7:00 PM - 8:15 PM). Integration with existing backend systems to store and manage biometric data securely.

3. System Requirements Specification:

3.1. Functional Requirements:

- User Management: Secure registration and login functionalities for administrators and staff members.
- Room Capacity Management: Interface for adjusting room capacities and integrating with existing data systems.
- Room Vacancy Status Tracking: Real-time updates on room occupancy statuses and dynamic visualization tools.

- Room Allocation Details: Comprehensive records of room allocations, accessible to authorized users.
- Type of Room Information: Catalog of room types, including specifications and amenities.
- Hostel Block Details: Overview of hostel infrastructure, with features for accessing block layouts and floor plans.
- Outsider Room Booking: Portal for outsiders to request room bookings, with approval workflows and payment processing.
- Total Admitted Strength: Analytics dashboard for tracking hostel admissions and occupancy trends.
- User Complaint System: Platform for students to report hostel issues, with administrators having the ability to respond.

3.2. Non-Functional Requirements:

- Performance: The system should respond to user actions promptly and handle concurrent user loads efficiently.
- Security: User data must be encrypted during transmission and storage, with access restricted to authorized users.
- Usability: The user interface should be intuitive and user-friendly, with clear error messages and guidance provided.
- Reliability: The system should be available 24/7 with minimal downtime, and data backup and recovery mechanisms should be in place.
- Scalability: The system should be designed to accommodate increasing user and data

volumes, with provisions for future feature enhancements.

Backend:

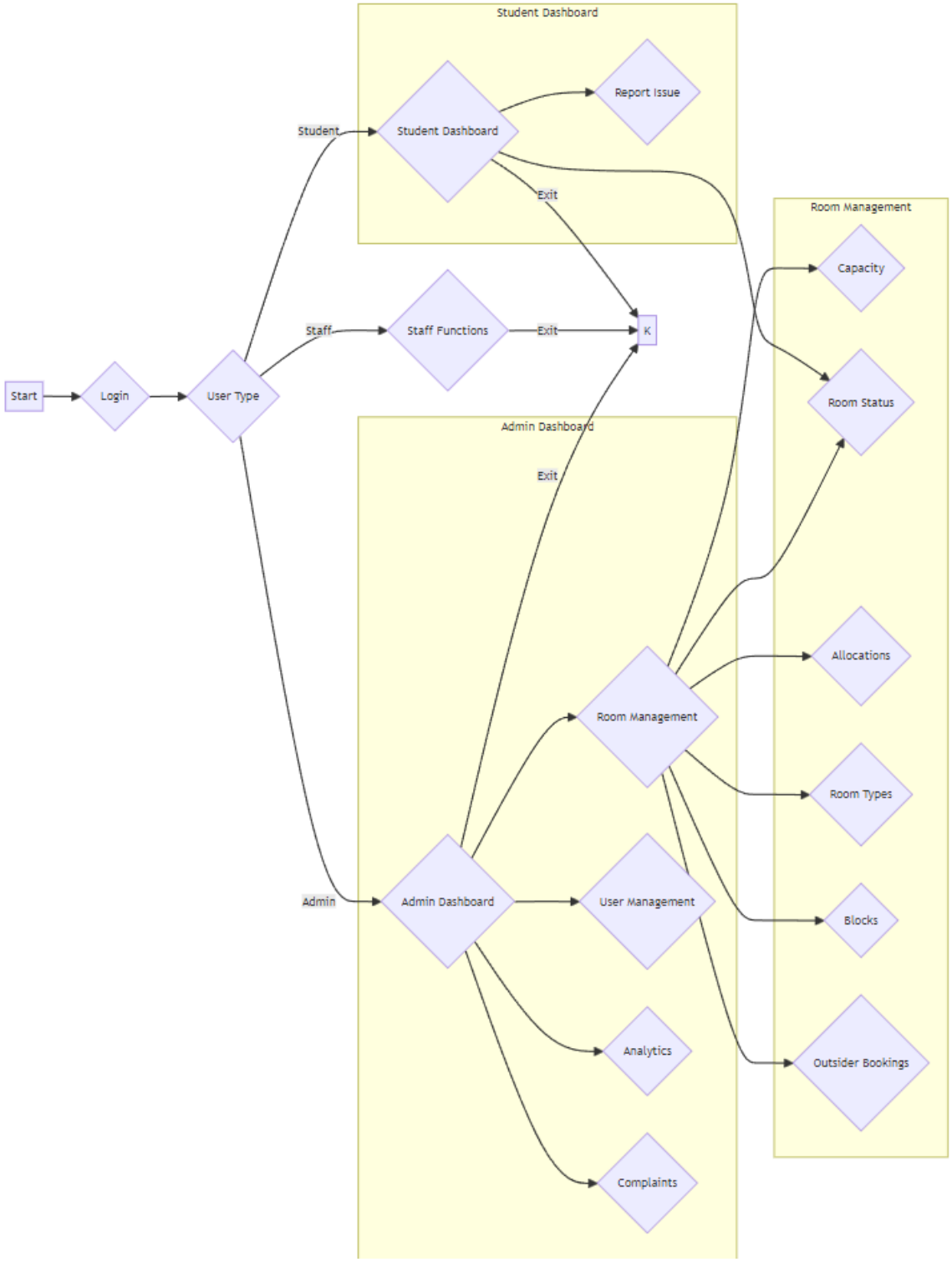
1. Student:
    - Attributes:
        - name: String
        - email: String
        - password: HashCode
        - rollNo: String
2. Room:
    - Attributes:
        - roomId: String
        - roomType: String
        - capacity: Number
        - currentOccupancy: Number
        - occupants: Array of Students
        - isVacant: Boolean
3. HostelBlock:
    - Attributes:
        - blockId: String
        - blockName: String

        - rooms: Array of Rooms
        - facilities: Array of Strings
        - floorPlan: String
4. Complaint:
    - Attributes:
        - complaintId: String
        - student: Student
        - description: String
        - status: String
        - createdAt: Date

Stack:

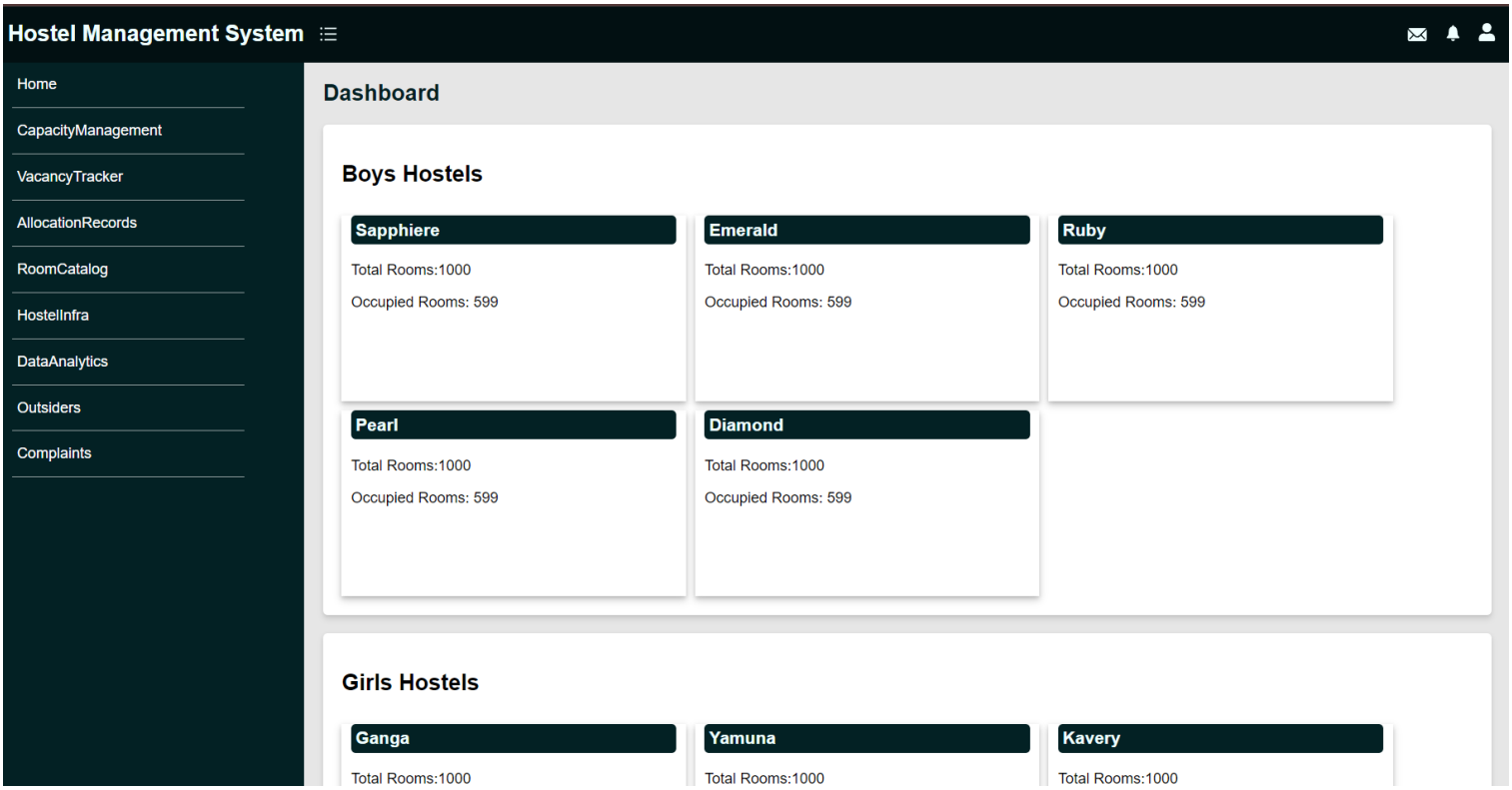
• Front End	React.js
• Back End	Node.js, Express.js
• Database	MongoDB

Flow chart:

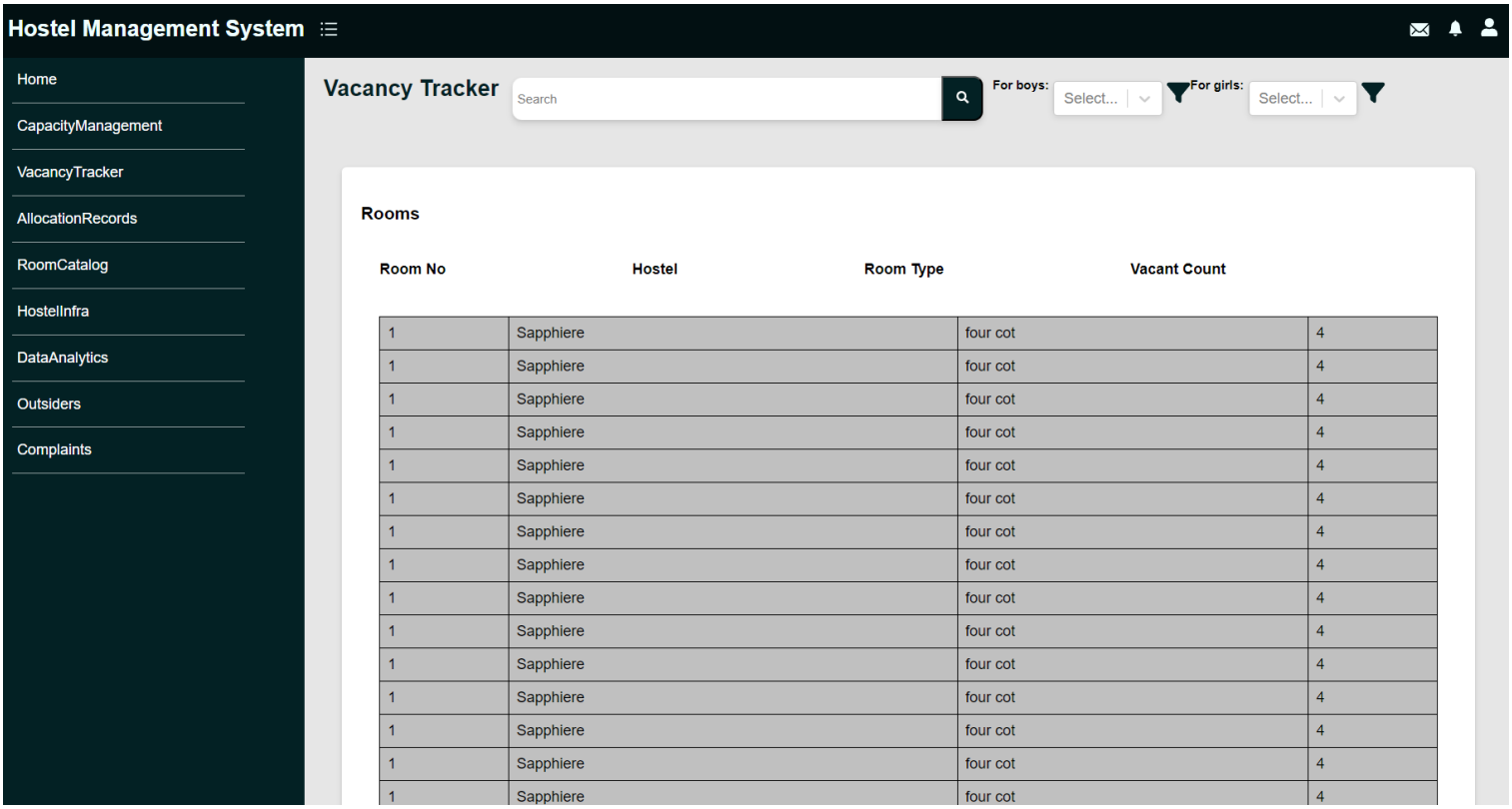

Student Dashboard
Report Issue
Student
Student Dashboard
Exit
Room Management
Capacity
Staff
Staff Functions
Exit
K
Start
Login
User Type
Room Status
Admin Dashboard
Exit
Allocations
Room Management
Room Types
Admin
Admin Dashboard
User Management
Blocks
Analytics
Outsider Bookings
Complaints

Prototype of the Project:

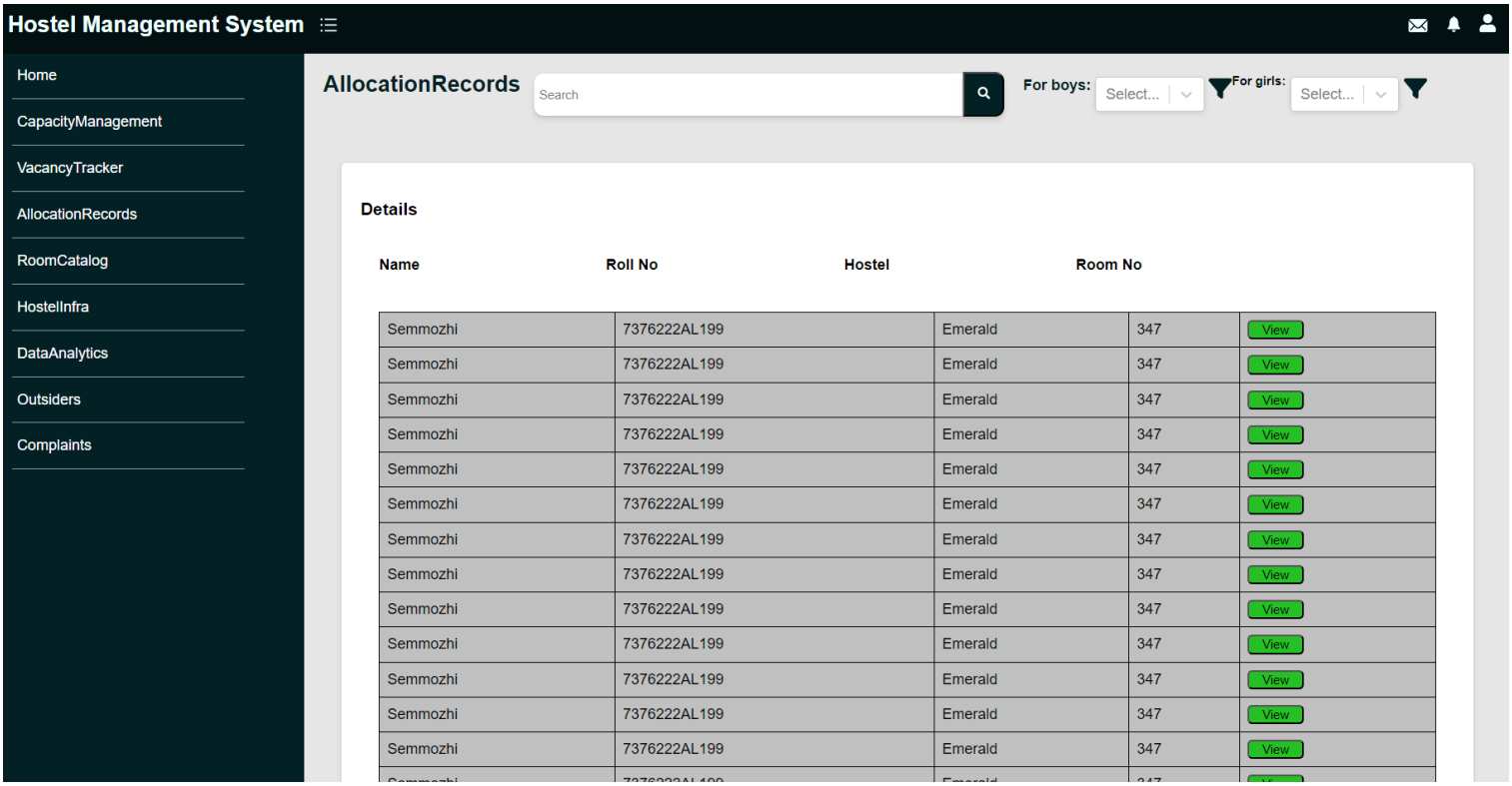
Dashboard



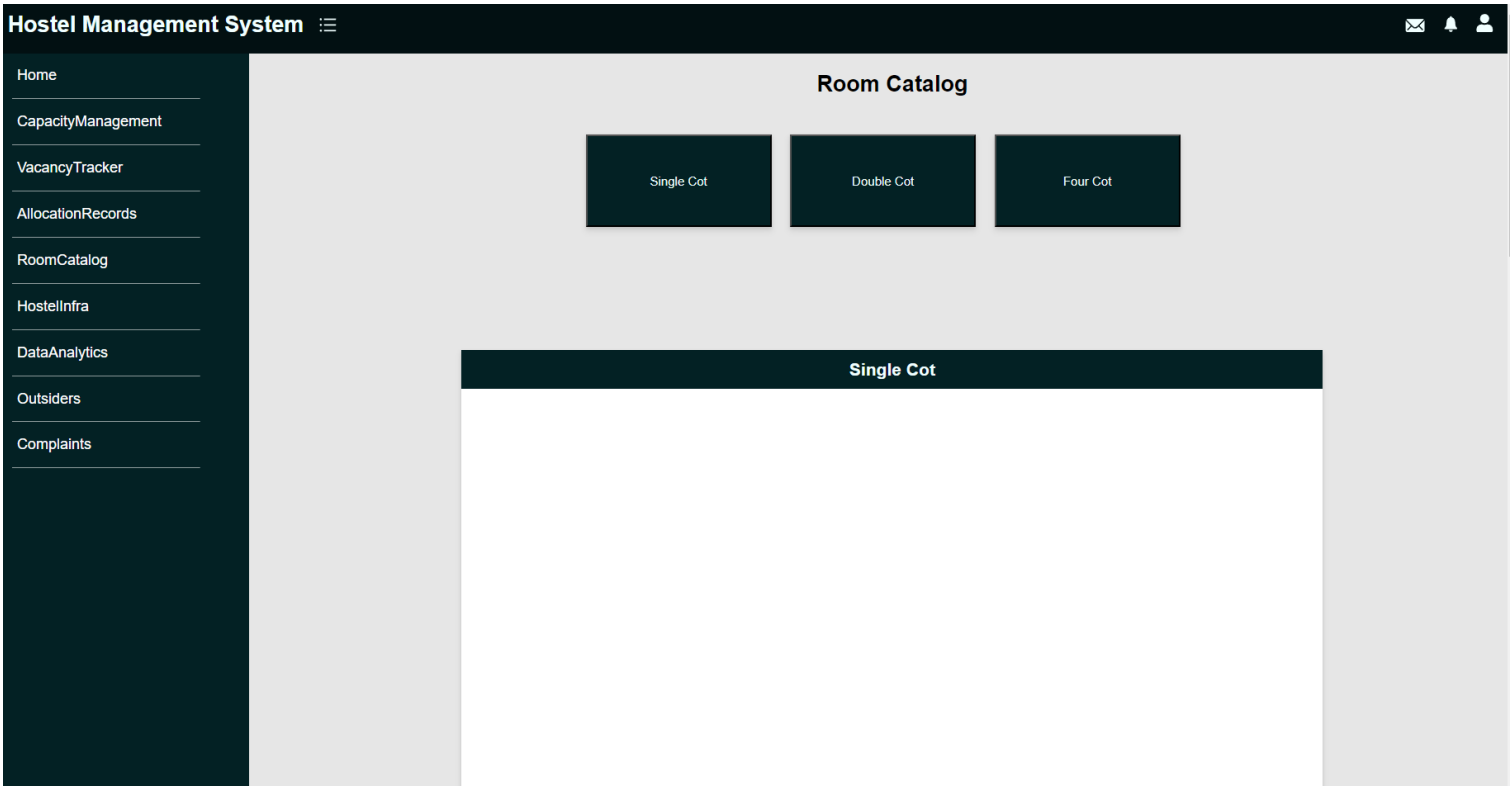
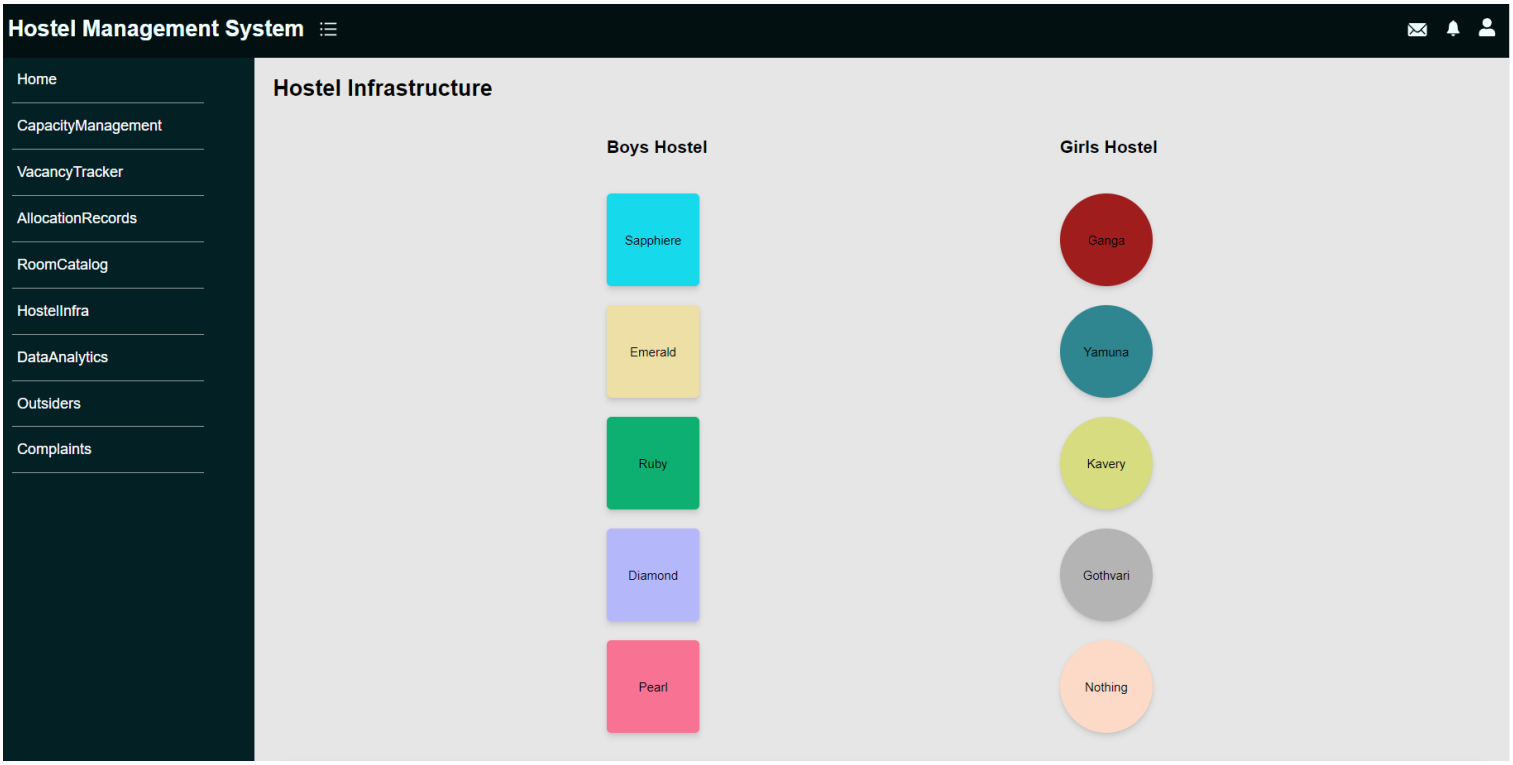
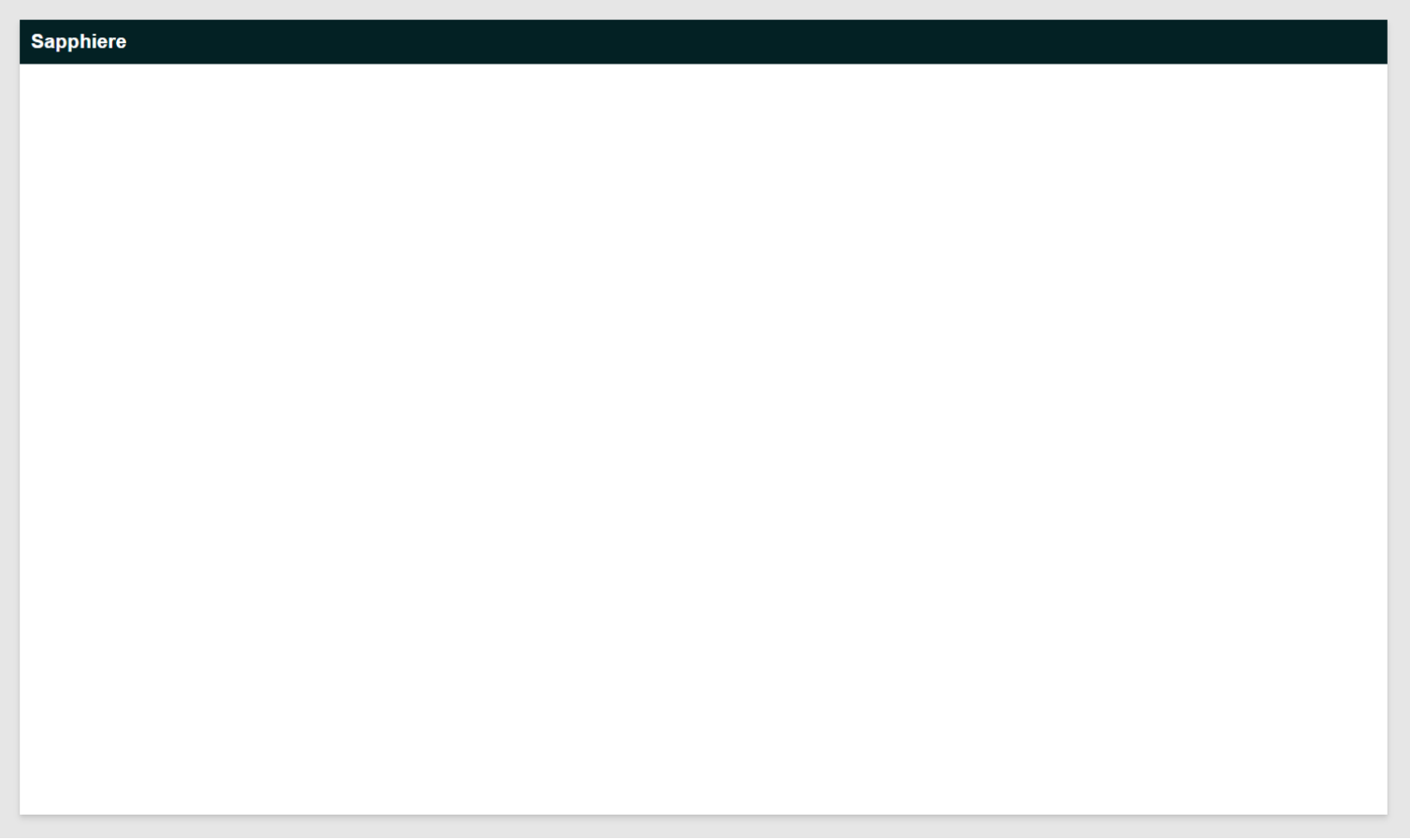
Vacancy Tracker



Allocation Records



Rooms Catalog



Hostel Infrastructure